# Cultivation Practices Of Fruits, Vegetables, Flowers And Medicinal Plants

## LAST 5 PREVIOUS YEAR QUESTION

**1. The sale of cut rose is maximum on which of the following days?**

a. Mother's Day b. rose day c. Valentine's Day d. all of the above

**2. What is the scientific name of jack fruit?**

a. Artocarpus heterophyllus
b. Malus
c. Fragaria × ananassa
d. None of the above

**3. What is the white powdery fungus that appear on leaves, stems, and sometimes fruit in grapes and hugely impact grape crop during the monsoon season?**

a. Damping off
b. Rhizopus
c. Downey mildew
d. Anthracnose

**4. What is the sensory and visual ripening index is generally preferred by the farmers to check the ripening of jack fruit.**

a. Rotten smell with pale yellow colouration
b. Fruity smell with change in colour from green to yellow
c. No smell while colour changes to yellow
d. None of the above

**5. Damping off is a very common disease in papaya while raising the seedlings in the nursery is caused by**

a. Phytophthora
b. Actinomycetes
c. Fusarium

d. Due to low water stress

**6. What is the scientific name of apple?**

a. Malus
b. Pisum stivum
c. Musa pudica
d. Psidium guajava

**7. Stalk end rot is a very common disease of which of the following fruit**

a. Banana
b. Apple
c. Grapes
d. Mango

**8. Name the disease of citrus where in the disease becomes noticeable by profuse gumminess is on the surface of affected bark which turns brown and develops longitudinal cracks. It also causes leaf-rot.**

a. Foot rot (gummosis)
b. Anthroconase
c. Damping off
d. Brown heart

**Answers**

**1.C 2.A 3.C 4.B 5.A 6.A 7.D 8.A**

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## 1. Mango

**Scientific name**-*Mangifera indica L.*

**Varieties-**Neelum, Bangalora, Alphonso, Rumani, Banganapalli, Kalepad, totapuri, sendura

- **Soil and Climate-**Red loamy soil with good drainage is preferable. pH range 6.5 to 8.
- **Season of planting** - July to December
- **Planting material** –Use plantable size grafts propagated through **inarching**, **soft wood** or epicotyl grafting.
- **Manures and fertilizers** may be applied during **September – October**, **45 – 90 cm** away from the trunk up to the peripheral leaf tip and incorporated.
- **Canopy management**-Remove **rootstock sprouts and low-lying** branches nearer to ground to facilitate easy cultural operations. **Remove overlapping, diseased**, **dried and weak branches in old trees** to get good sunlight and aeration.

**Figure 1 – Mango (Neelum)**

- **Carry out judicious pruning of the internal branches during August – September, once in three years.**
- **Growth regulators** -**Spray NAA @ 20 ppm at flowering to** increase the fruit set and retention
- **Pest disease**- Hoppers, Sooty mould, Mealy bug, Flower webber, Stem borer Fruit fly
- **Diseases-Powdery mildew, Anthracnose and Stalk end-rot, Sooty mould**
- **Treatment of disease**- spray a recommended solution of **carbendazim, Carbofuran, mancozeb, wettable sulphur to treat powdery mildew** etc.
- **Harvest Season**: March to June.

- **Post harvest treatment-Dip the fruits in 52° ± 1°C hot water** immediately after harvest for 5 minutes followed by 8% plant wax (**Fruitox or Waxol) to reduce anthracnose disease** in mango during storage. Two pre-harvest sprays of 0.2% Mancozeb (2.0 g / lit) will also reduce the incidence.

## 2. Banana

**Scientific Name**-*Musa sp*

**Varieties**-Grand Naine, Robusta, Dwarf Cavendish, Rasthali, Vayal vazhai, Poovan, Nendran, Red Banana, Karpooravalli,

**Figure 2 – Banana (Robusta Variety)**

**Soil and Climate**- Well drained loamy soils are suitable. Alkaline and saline soils should be avoided.

**Season of planting-** January – February and November – December and in hilly areas during august and September

**Planting material**- Suckers

**Special Practices**-**Bunch thinning** i.e., removal **of one to two small bottom hands** from the bunch (keeping only 7 to 8 hands) facilitates uniform bunch development and this practice is recommended for export purpose.

**Plant disease**

1. **Pest disease**- Rhizome weevil, Pseudo stem borer, Banana aphid, **Bunchy-top**, Thrips Lace wing bugs
2. **Diseases**-Sigatoka leaf spot, **Panama Disease (Fusarium wilt), Cigar end rot**

**Plant protection from disease – spray a recommended solution of carbendazim, Carbofuran, mancozeb etc.**

**Crop duration:** The bunches will be ready for **harvest after 12 to 15 months** of planting.

**Harvest**: Bunches **attain maturity from 100 to 150 days** after flowering depending on variety, soil, weather condition and altitude.

## 3. Acid Lime

**Scientific Name** - ***Citrus aurantifolia***

**Varieties** - PKM 1, VRM 1 Tahithi lime, Balaji (Tenali), Vikram, Rasraj, NRCC -7, NRCC -8, Phule Sharbathi

**Figure 3 – Acid Lime (Lemon)**

**Planting Material**- budded plant

**Soil and Climate** - **Areas with dry climate and moderate rainfall** (<750 mm) are best suited. **Deep well drained medium black**, loamy or alluvial soils rich in organic matter and devoid of calcium carbonate are good. The optimum **pH is 6.5 -7.0**

**Season:** December – February and June – September.

**Harvest**: Starts bearing from 3 - 4 years after planting.

**Post harvest handling**

Harvested fruits are **graded according to size and colour and** packed in bamboo baskets or wooden crates lined with green foliage. Treating the fruits **with 4% wax emulsion** followed by pre-packing in 200-gauge polythene bags with 1 % ventilation improves the shelf life for more than 10 days.

**Yield** - 25 t /ha /year.

**Pest disease** - Fruit sucking moth, Shoot borer, Fruit fly, Citrus root nematode etc.

**Disease**- Twig blight, Canker, Tristeza virus,

## 4. Sweet Orange

**Scientific Name** - ***Citrus sinensis***

**Varieties** - Sathugudi, Batavian, Mosambi, Malta blood red, Cutter Valencia.

**Soil and Climate**-**Deep well drained loamy soils** are the best for the cultivation of Citrus. pH of soil should **be 6.5 to 7.5**. A dry climate with a**bout 50 – 75 cm of rainfall from June – September** and with well-defined summer and winter season is ideal.

**Figure 4 – Malta blood red**

**Season:** July to September

**Harvest:** Starts **bearing fruit from 5th year after** planting. **Sweet orange takes 9-12 months for maturity**. Being non-climacteric fruits should be harvested only after full maturity.

**Yield: 30 t / ha.**

**Pest:** Leaf miner, Citrus root nematode

## 5. Grapes

**Scientific name** -*Vitis vinifera L*

**Varieties** - Muscat Hamburg, Pachadraksha, Anab-e-Shahi, Dil Kush, **Thompson Seedless, Red Globe**, Tas-A-Ganesh, Manik Chaman, Sonaka, **Sharad Seedless**, Nana Saheb Purple, Crimson Seedless, Fantasy Seedless, Italia, Flame Seedless and Clone A-18/3.

**Soil and Climate** - Grapes prefer **dry humid condition** for better growth and yield. Areas within the temperature range of **15 to 40$^0$ C** and rainfall lies **between 500 and 900 mm** are suitable for this crop.

- **Rain should not coincide with** vine growth after pruning and bunch ripening phase. Cloudy weather with high relative humidity, fog and low temperature are not suitable for flowering and fruit set.
- **This results in build-up of fungal diseases** (powdery mildew) which ultimately leads to loss in yield.
- Well-drained **rich loamy soil with a pH of 6.5 - 7.0**. The day time temperature ranging **from 20$^0$C to 35$^0$C is optimum for** proper vine growth and establishment.

**Figure 5 – Thompson Seedless Grape**

**Planting material**- **Rootstock** and the best time for planting root stock is **January – February**

**Field preparation and Planting**

The grapevine can **be trained over pandal system** or **improved Y trellis** training system.

**Pruning Season**

**Summer crop:** Pruning in December - January and harvesting during April - May.

**Winter crop**: Pruning in May-June and harvesting during August – September

**Special viticultural practices**- Shoot thinning, Training the shoots, Tipping, Nipping, Cluster and berry thinning

**Tipping is practised by the retention of 9-11 leaves above the last cluster and tying of clusters** in the pandal after the fruit set for berry development and maturity in the bunches. Further tendrils are also removed.

**Nipping is practised by removing the emerged shoots from axillary buds and terminal growth at 12th to 15th bud position from the base.**

**Plant Protection**

**Pest-** aphids, mealy bugs, Thrips

**Plant disease**- Powdery mildew, Downy mildew, Anthracnose, bacterial leaf spot

**Disease treatment**- wettable sulphur powder, Bordeaux mixture, mancozeb etc.

## 6. Guava

**Scientific name** - *Psidium guajava L.*

**Varieties** -**Allahabad Safeda**, Lucknow 49, Arka Amulya, Arka Mirdula, Banarasi, TRY (G) 1, **Arka Kiran**, Lucknow 46, Arka Reshmi, Lalit

**<u>Soil and Climate</u>** - Guava grows well **both in wet and dry regions** but it does better under irrigation in the dry tracts. It can be **grown up to 1000 m altitude**. Well drained soils are the best. **Tolerates salinity and alkalinity**. In saline soils, add 3 kg gypsum/plant during planting and once in three years after planting.

**Figure 6 – Allahabad Safeda Guava**

**<u>Propagation Materials/ Planting Material</u>** - Layers and grafts

**<u>Season Of Planting</u>** - June - December.

**<u>Harvest</u>**-Grafts/ layers **come to bearing within 2-3 years after planting First crop February – July, Second crop: September – January**.

**<u>Yield:</u>** 25 t / ha.

**<u>Important Disease</u>**

**Red rust**- to treat this **disease spray copper oxy chloride**

## 7. <u>Pineapple</u>

**<u>Scientific name</u>** - ***Ananas sativus*** **(Bromeliaceae)**

**Figure 7 – Kew Pineapple**

**<u>Varieties</u> - Kew**, Mauritius**, Queen**, Amritha and MD 2

**<u>Soil and Climate</u>** - **Mild tropical climate** as found in the **humid hill slopes** is best suited. It can be grown in plains under shade. Elevation from **500 m to 700 m** is ideal. A light well drained soil with **pH of 5.5 to 7.0 is preferable**. Heavy soils can also be used if drainage facilities are available.

**<u>Season:</u>** July – September

**<u>Crop duration</u>**: 18 – 24 months.

## 8. Papaya

**Scientific Name** - *Carica papaya L.*

**Varieties** - CO 2, CO 3, CO 4, CO 5, CO 6, CO 7, **Red Lady** is also being grown for commercial purposes.

**Soil and Climate** - It is a **tropical fruit** and grows well in regions where summer **temperature ranges from 35⁰ C – 38⁰ C**. Tolerates frost and comes up to an elevation of **1200 m**. Well drained soils of uniform texture are preferable. **If drainage is not adequate, collar - rot disease may occur.**

**Figure 8 – Red Lady Papaya**

**Sowing** - **500 g of seeds are required for planting one ha**. Nursery is prepared then **60 days old seedlings are transplanted** in field.

- After cultivation **Male plants should be removed** after the **emergence of inflorescence** maintaining **one male plant for every 20 female plants for proper fruit-set.**

**Crop Duration**: 24 – 30 months.

**Harvest**: Fruits should be **picked at colour break** stage.

**Papain production**- Papain has several industrial uses, the important one being in brewing industries. It is used as —meat tenderiser, and in textile and leather sanforization, processes and drugs. **Extracted from 75 to 90 days old fruit**.

**Disease**

**Root rot and wilt- in water stagnation areas** and Papaya ringspot disease

## 9. Pomegranate

**Scientific Name** - *Punica granatum L.*

**Varieties**- Jyothi, Ganesh, Arakta, **Rudhra, Mridhula**, Bhagwa, Ruby, YCD-1, G 137, Solapur Lal and **Solapur Anardana**, **Bacterial blight tolerant varieties**: Nayana, Kalpitiya, Nana and Daru.

Figure 9 – Mridhula Variety Of Pomegranate

**Soil and Climate** - It is grown in a **wide range of soils**; drought resistant and tolerant to salinity and alkalinity. **Cool winter and dry summer** are necessary for production of high-quality fruits. It performs **well up to1800 m** elevation.

**Planting Material -** Hard wood cutting

**Crop Practices** – Thinning, Pruning, Bahar treatment, Thinning of flowers etc

## 10. Jack

**Scientific Name** - *Artocarpus heterophyllus*

**Varieties** - Velipala, Singapore, Hybrid jack, Panruti selection, Thanjavur jack, Burliar 1, **PLR 1 and PLR (J) 2** and PPI 1.

**Propagation** -Softwood grafting approach grafting method: **Large scale propagation of jack can be done by cleft grafting during July - August on 4-month-old seedling rootstock**

**Harvest** - Yield **commences from 5th year in grafts** and 8th year in seedling trees. **Harvest during March-July.**

Figure 10 – PLR 1 Jackfruit

## 11.Ber

Figure 11 – Ber

**Scientific Name** - *Zizyphus mauritiana Lam*

**Varieties -** Goma Kirti, Thar Sevika, Thar Bhubhraj, Kaithali, Umran, Gola and Banarasi.

**Soil and climate** - The ber plant comes **under arid and semi-arid** situation. Tolerates salinity and alkalinity.

**Planting Material** – **T budding and patch budding**

**Diseases** - **Black leaf spot**, **Powdery mildew**

## 12. Amla/ Aonla

**Scientific Name** - *Phyllanthus emblica*

Figure 12 – Kanchan variety

**Varieties** - BSR 1, Goma Aiswarya, Banarasi, NA 7, Krishna, **Kanchan**, Chakaiya,

**Soil and Climate** - Amla is a **sub-tropical plant** and prefers dry climate. Hardy plant, it can be grown in varied soil conditions. Tolerates salinity and alkalinity

**Planting material** - **Softwood grafting and patch budding**

**Diseases**- **Ring Rust** or Aonla Rust, Fruit Rot / Bird's eye spot

## 13. Apple

**Scientific Name** - *Malus domestica*

**Soil and Climate** -**Red lateritic soils** with good drainage and high organic matter are more suitable. The soil pH should be around **5.8 to 6.2**. **Liming is essential** to correct the pH of acidic soil. A **minimum depth of 1.6 metre** is desirable.

**Planting material** -**One year old grafts on M.778** and M.779 rootstocks during **June – July.**

**Figure 13 – Ruby apple**

**Season** -June to December

Training and Pruning **The tree is trained to open centre system**. **Prune the tree every year during the month of December – January.**

**Disease**

**Pest**-Wooly aphids, Stem borer, San jose scale

**Disease- Scab**, Powdery mildew, premature leaf fall,

## Principles Of Vegetable Cultivation

As compared to orcharding, the vegetable industry is characterized by its flexibility. Because most vegetables are grown as annuals, shifts in cultivars and crops can be readily made.

**There are three main categories of vegetable production: home gardening, market gardening, and truck gardening**

**Home gardening**

Home gardening means the cultivation of flowers, fruits, vegetables, or ornamental plants for **personal use of the owner or tenants** of a lot.

**Market Gardening**

Market gardening specializes in the **cultivation of high value perishable crops such as vegetables, fruits and flowers, solely for the urban markets**. They are situated close to urban markets mainly because of the high value crops.

**Truck Farming**

Truck farming horticultural practice **of growing one or more vegetable crops on** a large scale for shipment to distant markets. It is **usually less intensive and diversified than market gardening**. At first this type of farming depended entirely on local or regional markets.

**<u>vegetable crops have also been classified based on method of cultivation.</u>**

In this method all the crops, which have similar 24 cultural requirements are grouped together.

1. **Cucurbits** (musk melon, water melon, bottle gourd, ridge gourd, cucumber etc.),
2. **Cole crops** (cauliflower, cabbage, broccoli),
3. **Root crops** (beet root, carrot, radish),
4. **Bulb crops** (onion and garlic)

**There are some principles required in the production of vegetable crops which are very important and well known to the grower**.

**These principles are:**

- Production of **vegetables does not involve a long-time investment** as does in the orchard of guava, mango, or apple.
- Vegetable growers/farmers **are not bound to produce the same crop each** year like his counterparts, who grow fruit crops.
- Vegetable growing **lacks the stability which is methodically** developed over a period of years like an orchard thus, getting into vegetable production is a fast process and getting out may even be faster.
- Vegetables can be grown by people with limited experience. Only skilful farmers sustain their vegetable production.
- The land for production of **vegetable crops is flexible and adjustable**. It is much easier for vegetable growers/farmers to change production from one crop to another than for fruit crop grower.
- **Cooperative efforts and organizations are somewhat more difficult with vegetable crop producers than fruit growers**. Vegetable/grower/farmers have no long period for making plans.
- Vegetable **production is seasonal**. Vegetable production requires more intensive production management per unit area and time.

**<u>Vegetable Nursery Raising</u>**

Most vegetable species **are grown from seeds**, but some important ones (e.g. pointed gourd, colocasia, basella, ivy gourd etc.) are **propagated by vegetative methods**. Among those grown from seeds, a significant number mainly those with small seeds (e.g. tomato, brinjal, cauliflower, cabbage etc.) **are usually first sown in nursery beds**, boxes or containers and are transplanted at a later stage.

**Nursery raising have several advantages** like economy of seeds, uniformity of growth and selection of vigorous & healthy seedlings for transplanting

**Field Establishment**

**Land Preparation**: In preparing land for vegetable production, factors such **as ecological location, mode of cropping, season, crop disposition, and the type of vegetables to** be grown should be taken into consideration. The land should be first cleared off of existing vegetation followed by levelling and suitable tillage operations

**Planting**

Vegetables can be **propagated either by direct sowing or by transplanting methods**

**Direct Sowing**

Vegetables are sowed **either by broadcasting or by seed drilling methods**.

- In **broadcasting, seeds are spread over the prepared land by throwing in small quantities example Celosia and Amaranthus**.
- **Seed drilling method** is followed for **planting small seeded vegetables in rows**. Shallow furrows are made at the spacing recommended for the crop and the seed drilled along the furrows. This method can also be used for some leafy vegetables, carrot, radish etc

**Transplanting method**

Vegetable **seedlings are first raised in the nursery** for a required period of time before they are transplanted on the field. Seedlings are **transplanted in the morning or** in the evening to avoid transplanting shock.

**Some Routine Operations**

**Thinning:** Thinning of vegetable is done to reduce the number of seedlings per stand. Supplying or gap filling: This is the practice of providing missing stands of vegetables planted by direct sowing as a result of poor emergence or when seedlings are damaged by pests.

**Staking:** This is usually required for **vegetables with climbing growth habit such** as fluted pumpkin, or those **with weak stems such as tomato**. Stake can be made from bamboo or other available wood. The support allows the plant to carry more load without touching the soil thus enhancing the quality of the fruit.

**Mulching**: A mulch is a layer of plant residue or other materials like plastic or paper, which is applied to the surface of the soil in order to reduce evaporation, run-off or to prevent weed growth. The purpose of mulching is to conserve soil moisture.

**Watering**: Young vegetable seedlings in the nursery or in the field **should be watered in the early morning or in the evening**. Watering should be done before transplanting particularly in the evening. Likewise, over-watering can be very harmful and can encourage the development of pathogenic diseases and also cause mechanical damage to the seedlings.

**Fertilizer application**: Vegetables must be provided with ample supplies of nutrients such as nitrogen. Application **of N fertilizer has been shown to increase yield**. In some tropical leafy vegetables, **fertilizers such as FYM and other sources of P and K** can be applied as pre-plant basal dressing or after the plants have become established as post planting application.

**Weeding:** Weeds can be managed **using cultural, physical, chemical and biological methods.** Weed seeds and rhizomes can be killed using physical method during land preparation by burning. Mulching of soil can also smother weeds. **Hoeing, pulling and roguing are carried out during the early stages of growth.** Chemical weed control is applied in commercially grown vegetable crops

## Cultivation Practices Of Vegetables

### 1. Tomato

**Scientific Name** - *Solanum lycopersicum L.*

**Varieties** - PKM 1, CO 3 (Marutham) and Paiyur 1

**Hybrids** -CO 3, Arka Rakshak, Arka Samrat

**Soil - Well drained loamy soil rich** in organic matter with a pH range of 6.5 - 7.5.

**Season of sowing** - May - June and November – December

**Figure 14 – Arka rakshak**

**Seed Rate - Normal Varieties** 300-350 g / ha, **Hybrids**: 100-150 g / ha

**Plant Disease**

**Pest Disease** -Fruit borer, thrips, pin worm

**Disease**- Damping off, Leaf spot, Tomato early blight, **Fusarial wilt and root knot nematode**

**Duration of crop** -110- 115 days from transplanting (135 - 140 days from sowing)

**Yield - Varieties** 30-40 t / ha, **Hybrids:** 80-95 t / ha

**Spray of bloom flower – N** @ 2ml / litre at – 30-, 55- and 75-days planting increase the yield.

## 2. Potato

**Scientific Name -** *Solanum tuberosum L.)*

**Varieties -** Kufri Gridari, Kufri Jyoti, Kufri Muthu, Kufri Swarna, Kufri Thangam and Kufri Malar

**Figure 15 – Kufri Jyoti**

**Soil And Climate**

The soil should be friable, porous and well drained. The optimum **pH range is 4.8 to 5.4**. It is a cool weather crop. Potato is mostly grown as a rainfed crop. Cultivated in regions receiving **a rainfall of 1200 - 2000** mm per annum.

**Season and planting**

**Hills** -Summer: March – April, Autumn: August – September, Irrigated: January – February

**Plains** - October – November.

Use **disease free, well sprouted seed tubers weighing 40 – 50 g.** Use **Carbon disulphide 30 g/100 kg of tubers for breaking the dormancy** and inducing sprouting of tubers. Plant the tubers at 20 cm apart.

**Seed rate** -3,000 – 3,500 kg/ha.

**Preparation of field Prepare the land to fine tilth**. Provide drainage channel along the inner edge of the terrace. Form **ridges and furrows with a spacing of 45 cm** between ridges.

**Irrigation** - Irrigate the **crop 10 days after** planting. Subsequent irrigation should be given **once in a week**.

**Plant protection**

**Pest-** Cut worms, Potato tuber moth, Aphids, Jassids

**Other Diseases**- **Late blight, Brown rot, Early blight**, Virus diseases

**Yield:** 15 – 20 t/ha in a duration of 120 days

## 3. Onion - Small onion

**Scientific Name** - *Allium cepa*

**Varieties** - CO 1, CO 2, CO 3, CO 4 and MDU 1, **CO (On) 5 is a free flowering and seed setting type.**

**Soil**

**Red loam to black soils** with good drainage facilities are ideal. The germination and bulb maturation are affected in clayey soil. It grows **well in pH range of 6- 7 and** a mild season without extremes of heat and cold.

**Figure 16 – CO K2 Onion Variety**

**Season And Sowing**

**Plant the medium sized bulbs** during **April – May and October – November**.

It requires sufficient soil moisture during its growing period **but heavy rains during bulb sprouting** and bulb formation affects the crop growth.

**Seed rate -Seed bulbs 1000 kg/ha**. Medium sized bulbs are to be chosen for planting. **Seeds @ 2.5 kg/ha.**

**Preparation Of Field-** Plough the land four times to a fine tilth. **Form ridges and furrows at 45 cm spacing**. Plant the bulbs or seedlings on both the sides of the ridges at 10 cm apart.

**Irrigation-**Irrigate at the time of planting of seedlings and third day and later at weekly intervals. **Withhold irrigation 10 days before harvest**.

**Plant disease**

**Pests-Thrips**, Onion fly, Leaf spot / **purple blotch**, Basal rot,

## 4. Big Onion Or Common Onion

**Scientific Name** - Allium cepa

**Varieties-** Bellary Red, Pusa Red, NP 53, **Arka Niketan, Arka Kalyan**, Agri Found Light Red and Agri Found Dark Red

**Figure 17 - Arka Kalyan Onion**

**Soil -Red loam to black soils** with good drainage facilities. The optimum pH is 5.8-6.5.

**Season** -May – June. Mild season is preferred

**Seed rate and Sowing:** 7.5 kg/ha.

**Preparation of main field:** Plough the land to a fine tilth. Form **ridges and furrows at 45 cm spacing**. Plant 45 days old seedlings at 10 cm apart on both the sides of the ridges.

**Irrigation** Irrigate at planting and **on third day and** later at weekly intervals.

**Plant disease -**

**Pest** -Thrips, Root grub,

**Disease-** Leaf spot / purple blotch

### 5. Chillies

**Scientific Name** -*Capsicum annuum*

**Varieties**-K 1, K 2, CO 1, CO 2, CO 3, CO 4 (vegetable type), PKM 1

**Figure 18 – K2 Chili Variety**

**Soil: Well drained loamy soil rich** in organic matter with pH range 6.5-7.5.

**Season Of Sowing** -January – February, June – July and September – October

**Seed Rate** -Varieties-1.0 kg / ha, Hybrids -200 - 250 g / ha

**Nursery Area**: 100 sq.m / ha.

**Field Preparation** -Thoroughly prepare the field with the addition **of FYM @ 25 t/ ha** and form ridges and **furrows at a spacing of 60 cm**.

**Irrigate the furrows and transplant 40-45 days old seedlings**, with the ball of earth on the ridges.

**Spacing-** Varieties: 60 x 45 cm, Hybrids: 75 x 60 cm

**Irrigation** -Irrigate at weekly intervals.

**Plant protection from**

Fruit borer, Thrips, Aphids, Yellow Muranai mite,

**Diseases-Damping off**, Leaf spot, Powdery mildew, Anthracnose**, Die-back** and fruit rot, Chilli mosaic, **Root knot nematode**

### 6. Cauliflower

**Scientific Name** - *Brassica oleracea*

**Varieties** -**Hills** Ooty 1, **Pusa Deepali**, **Plains** -Early Synthetic, Pawas

**Climate and Soil** -It requires **cool moist climate**. Deep loamy soils are good with high organic matter and good drainage. It can be grown in a **pH range of 5.5 to 6.6.**

**Season And Sowing**

The early varieties **may tolerate higher temperature and long days**. This can be grown in **plains during September to February**. Late Varieties Snowball types can be grown in hills.

**Seed Rate -375 g/ha.** Sow the seeds in **raised beds and transplant 25 days** (early varieties), **45 days (late varieties**) old seedlings at 45 cm apart.

**Preparation Of Field**

Bring the soil to fine tilth. **Pits should be taken at a spacing of 45 cm** either way in hills. Form ridges and furrows at 60 cm in plains.

**Figure 19 – Pusa Deepali Cauliflower**

**Irrigation**

**Hills: Once in a week** during January and February.

**Plains**: Once in a week.

**Plant pest**-Aphids, Diamondback moth,

**Diseases**- **Club root disease**, Leaf spot, Leaf blight,

**Physiological disorders**-

- **Browning or brown rot**- This is caused by boron deficiency.
- **Whip tail** - results from the deficiency of molybdenum.
- **Buttoning** -The term buttoning is applied to the **development of small curds** or buttons.
- **Blindness** -Blind-cauliflower plants **are those without terminal buds**. The leaves are large, thick, leathery and dark green.

**Yield -Hills**: 20 – 30 t/ha, **Plains**: 15 – 20 t/ha

## Principles Of Flower Cultivation

In a garden there are certain operations that are to be followed judiciously for successful cultivation of ornamental plants. **Most of these operations, such as transplanting, pruning, pinching, etc., are of vital importance for the growth of the plants**. Some other operations, such as topiary or shearing of hedges, are designed to enhance the ornamental or aesthetic value of the plant material.

**Some routine operations**

**Soil Sterilization:** This is an essential operation to **eliminate soil-borne diseases**, caused mainly by fungi. Soil can be sterilized with 2 per cent formalin (or formaldehyde). The formalin solution is mixed thoroughly with the soil and this is covered with tarpaulin or gunny bag for 48 hours.

**Seed Sowing**: Sowing of seed is an important operation for annuals, biennials, and some herbaceous perennials, **as seeds of most of them are very minute and need special** care during sowing.

**Pricking:** **Pricking out, also known as thinning out, means removing the seedlings** from their original container and replanting into individual pots or transferring into beds to give them more growing space.

**Planting and Transplanting**: Planting and transplanting are two important operations. The time of planting depends on the climate of the area. **Deciduous plants are transplanted during the dormant season when they are in leafless conditio**n. **Under Indian conditions, the best period is to transplant during the rainy season**, provided there is no water-logging. The pits should be dug before the rains start.

**Transplanting:** There are **generally four types of transplanting in a garden**:

(1) Transplanting **a potted plant to ground**,

(2) Transferring a **potted plant to another pot**,

(3) **Potting a ground plant**,

(4) Transplanting **of large trees and shrubs** from the ground to another suitable location poses problems. For lifting very large trees, the help of tree-lifters or cranes and trucks will be needed.

**Shading**: Shading and protection of plants is an important operation in a garden, especially in places having hot summers and severe winters. **The newly planted seedlings or plants need shade from the scorching summer sun**. In places, where frost is likely to kill the plants, these should be given proper protection. The seedlings in the nursery also need shading

**Stopping or Pinching**: The operation of **pinching or stopping involves the removal of the growing-point of a shoot along with a few leave**s. **The two main purposes of this operation are to encourage branching to produce a bushy growth,** and/or the **production of flower-buds on the branch which is pinched**. Pinching is done mostly in annuals and herbaceous perennials and is hardly required for any flowering trees. The plants which need **pinching include dahlia, chrysanthemum, carnation, brachycome** and marigold.

**Deshooting: Deshooting involves the removal of shoots that are not wanted**. Some flowering annuals and herbaceous perennials produce numerous side shoots and if all of them are allowed to flower, the size and quality of the flowers will be greatly reduced. **Deshooting of carnations grown for cut flower trade** and chrysanthemum for exhibition purposes are common practice.

**Defoliation**: The **removal of foliage** is known as defoliation. This is done mainly with a view to induce flowering in certain plants. Sometimes, **this can also be done to reduce transpiration loss during periods of moisture stress** and also during transportation of certain plants such as roses. In **jasmine, it is a common practice to defoliate the plants after pruning just prior to the flowering season**.

**Staking**: Plants in the **garden, either in pots or on ground, need support at least for** a part of, or throughout, its life. Stakes may be of various kinds. The most common stakes used in India are made of either whole bamboo or split bamboo of various sizes depending upon the type of plants to be staked.

**Pruning**: The **planned removal of branches, twigs, limbs, shoots, or roots is termed** as pruning. Even the removal of a dried flower can be termed as pruning. Each pruning is done with a view to increase the usefulness of a plant.

**Wintering:** This may be considered as an alternative to root pruning. In hotter parts of India, it may not be wise to resort to root pruning. **In such places ornamental plants are "wintered". During resting period, the water supply to the plant to be wintered is stopped for a few days and the roots are exposed to the sun by removing the surface soil around the trunk.** After wintering, the roots are covered with the same soil enriched with farm yard manure and copiously watered.

**Clipping or Cutting of Hedges and Edges**: To keep the hedges in symmetry, good health, and beauty, constant vigilance and regular shearing or pruning are needed.

**Topiary**: The art **of clipping and shearing shrubs and small** trees and sometimes even herbaceous perennials **into ornamental or abstract shapes is known as topiary**.

**Important Flower Crops**

| Flower | Scientific Name | Imp Practice | Important Verities |
|---|---|---|---|
| **Hybrid Rose (Cut Rose )** | *Rosa Hybrida* | Propagated By Rooted Cuttings Or By Budding Temperature With A Minimum Of 15°C And Maximum Of 28°C | Gladiator, Baby Pink, Sofia Lawrence, Ycd 1, Ycd 2, Ycd 3 |
| **Chrysanthemum** | *Dendranthema Grandiflora* | Terminal Cuttings Are Widely Used For Temp- 16 To $25^0$ C<br>Ph -5.5 To 6.5commercial Cultivation. | Bonfire Orange, Bonfire Yellow. |
| **Carnation** | *Dianthus Spp* | Temperature Of 18-$24^0$c<br>Well Drained Red Loamy Soil with Ph Of 5.5 - 6.5. | |
| **Gladiolus** | *Gladiolus Spp* | Temperature Range Of 27 - 30°C | |

| | | | |
|---|---|---|---|
| | | Sandy Loam Soil Rich in Organic Matter With Ph Of 6 To 7. | |
| **Rose** | *Rosa Sp* | Well-Drained Sandy Loam With Ph Of 6-7 Is Suitable | Edward Rose, Andra Red Rose, Button Rose Are Found Mainly Under Cultivation. |
| **Marigold** | *Tagetes Erecta L* | Drained Loamy Soil Is Found Suitable. The Soil Ph Should Be 7.0 To 7.5. | |

## Plantation Crop

### 1. Tea

**Scientific Name** - *Camellia sinensis*

**Soil and climate -**Tea requires **well drained soil with** high amount of organic matter and **pH 4.5 to 5.5. The** performance of tea is excellent at elevations ranging from 1000- 2500 m. **Optimum temperature is 20-27° C.**

**Figure 20 – Nipping of tea leaves**

**Preparation of cuttings-**Cuttings are **taken on April - May and August - September**. **Semi hardwood cuttings** are prepared with one full leaf and an internode with a slanting cut at the bottom.

### 2. Coffee

**Scientific Name** - *Coffea arabica L* (Rubiaceae)

**Soil** -Soil should **be deep, friable, open textured** rich in plant nutrients with plenty of humus and of **slightly acidic nature (pH – 4.5 to 6.5).**

**Propagation** -By seeds

**India produces two types of coffee: Arabica and Robusta**. Arabica has high market value than Robusta coffee due to its mild aromatic flavour.

**Figure 21 – Arabica Coffee**

## 3. Coconut

**Scientific Name** - *Cocos nucifera L.*

**Soil-** Light **sandy soils to heavy soils with a pH - 5.2 to 8.0**. Proper drainage, good water-holding capacity, presence of water table within 3 m and absence of rock or any hard substratum within 2 m of the surface.

**Rainfall:** 200 cm per year.

**Planting material** – seed

**Planting seasons** -June - July, December - January

**Figure 22 – Coconut**

## 4. Oil palm

**Figure 23 – Oil Palm**

**Scientific Name** - *Elaeis guineensis*

**Climate***:* Temperature is **21$^0$ C to 32$^0$ C**, annual rainfall - 200 cm and relative humidity - 75 – 100 %. Altitude – 450 -900 m.

**Soil**: Moist deep, loamy soils, rich in humus with good water permeability are suitable. Soil pH - 4 – 6

**Planting Materials**- seeds are used to prepare nursery bed

**Harvesting:** First harvest can be **done 3.5 to 4 years** after planting. Few ripe fruits are loose/fall off indicates the bunch is ready for harvest.

**Yield:** 25 – 30 tonnes of fresh fruit bunches/ hectare.

## Medicinal Plants

### 1. Tulsi

**Scientific Name** - *Ocimum Sanctum L*

Ocimum sanctum is native to India, **where it enjoys a religious attachment** and liked to be grown in shrines and homes as an aromatic perennial shrub.

**Tulsi leaves contain a bright yellow volatile oil which** is useful against insects and bacterial. The **principal constituents of this oil are eugenol, eugenol** methyl ether and carvacrol.

**Uses:** Tulsi **is an aromatic medicinal plant** is often taken in combination with other herbs. The fragrant leaves and flowers, in the form of tincture, tea or decoction are considered to be stomachic and expectorant, **used in treating coughs, bronchitis, skin diseases, and diarrhoea.**

**Figure 24 – Tulsi**

### 2. Aloe

**Scientific Name** - *Aloe barbadensis*

**Soil and Climate**-Well drained laterite to loamy soils is suited for aloe cultivation. The soil **pH must be ranged from 7.0 to 8.5**. Commercial cultivation can be done in regions having 25 – 40°C.

The planting can be done **during two seasons namely June – July** and **September – October.**

**Uses**- Aloe vera is a medicinal plant **with antioxidant and antibacterial** properties. **Aloe vera benefits can include reducing dental plaque**, accelerating wound healing, preventing wrinkles, and managing blood sugar.

**Figure 25 - Aloe Vera**

## 3. Stevia

**Scientific Name** - *Stevia rebaudiana*

**Uses - Calorie free sweetener**, sugar substitute for diabetic patients

**Economic Part**- Leaf

**Propagation**-Stem **cuttings of 15 cm length taken** from leaf axils of the current year's growth have been given better results. The best months for propagation are February-March. The cuttings will be ready for transplanting after **25-30 days of rooting**.

**Figure 26 – Stevia**

## 4. Lemongrass

**Scientific Name** - *Cymbopogon flexuosus*

**Soil And Climate**-Sandy loam with abundant **organic matter with a pH of 6.0** is found suitable. It comes **up well under tropical and sub-tropical** conditions with a high rainfall **(200 - 250 cm) and humidity**. It performs better up to 1500m above mean sea level.

**Seeds And Planting**-It can **be propagated by slips**. It can also be propagated through seeds at 4 kg/ha. Seedlings are to be raised and transplanted during June – July or October – November.

**Figure 27 – Lemon Grass**

**Uses -** Lemongrass is **used for treating digestive tract spasms**, **stomach ache, high blood pressure, convulsions, pain, vomiting, cough, achy joints (rheumatism),** fever, the common cold, and exhaustion. It is also used to kill germs and as a mild astringent. Used to make mosquito repellent creams. Commonly used as "lemon" flavouring in herbal teas.

## 5. Mint

**Scientific Name** - *Mentha sp*

**Figure 28 – Mint**

**Soil And Climate** - Well drained loamy soil high in organic matter content with pH 6.5 to 7.0 is suitable. Sub-tropical areas receiving an annual rainfall of 100 - 150 cm are good.

**Propagation** - It can be **propagated through root suckers and terminal cuttings**. Terminal cuttings are highly capable for rooting. The **root suckers** from the previous crop can be used for propagation.